

Ensuring Privacy and Accuracy in AI-Enhanced Education: A Proposal for
Customized ChatGPT implementations in the Classroom

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1. Abstract

This paper explores the capabilities and applications of ChatGPT, an AI-powered chatbot capable of generating human-like responses by utilizing reinforcement learning techniques. Although ChatGPT holds unlimited potential in the education sector, it is essential to address its security and privacy implications towards the intellectual property of educators and institutions. By exploring different approaches and discussions, this proposal examines the potential risks of student's current usage of ChatGPT and offers a solution to the current privacy problem educational institutions are facing while balancing it with the unlimited potential of generative AI. Finally, we analyze the open problems and limitations of the proposed solution to ensure the well-roundedness of our solution.

2. Introduction

In the education sector, recent development of artificial intelligence (AI) tools has been nothing short of revolutionary. Most notably, the rise of generative AIs (GenAI) like OpenAI's ChatGPT in December 2022 have emerged as powerful tools, offering not only students, but educators alike unlimited interactive and responsive support that can significantly enhance their educational experience. For example, in engineering and computer science, students can prompt ChatGPT to produce code with line-by-line comments, or to explain difficult concepts very clearly[2]. This is a major upgrade from the traditional chatbot that was developed, as with ChatGPT's natural language processing capability, it fixes the issue of chatbots being chatbots, as NLPs are much more friendly to users[4]. However, this technological breakthrough comes with its own set of challenges, most notably the concern for privacy, accuracy and the protection of intellectual property[2]. As students can now upload universities' course materials to publicly accessible AI platforms using GPT-4, the sanctity of academic content, a cornerstone of educational institutions, is at risk of erosion unless new measures are adopted. This proposal aims to address the privacy concerns without sacrificing the innovation, outlining a strategic approach for implementing a customized version of ChatGPT that operates within the secure boundaries of Concordia's digital infrastructure. We will explore how a customized generative AI can not only enrich the learning environment but also ensure that the educational materials that are crafted over countless hours by dedicated professionals remain within the infrastructure of their universities.

3. Background

In today's digital landscape, students and educators alike have easy access to Large Language Models and are ensuring their trust in them by providing it with potentially sensitive data concerning academic material. As artificial intelligence continues to be adopted into academic institutions, the preservation of sensitive data continues to be a significant worry, particularly as the potential of data breaches and data leak could compromise academic integrity and the intellectual work of professors and institutions can be at risk. To protect confidential data while increasing the learning rate, we propose implementing a private Large Language Model, named InternalGPT, that runs on algorithms derived on data provided exclusively by the institution itself to mitigate involuntary storage of data.

4. Problem

While AI promises a total technology transformation in higher education, privacy is a key challenge that is yet to be addressed in applying AI in education, especially with ChatGPT-4's ability to receive documents and pictures from users, according to OpenAI[9]. Specialists and researchers are heavily concerned with ethical considerations, especially with regard to data privacy and security. Zhang and Aslan bring attention to concerns about the critical flaws and limitations of the current AI ethics guidelines, which, according to them, can heavily impact research in the future [5]. Following up on this point, recent semi-systematic evaluation of 22 AI ethics guidelines also confirms Zhang and Aslan's statement by revealing that current guidelines have severe flaws and a range of AI ethics that are critical for AI research, development and implementation are actually missing or overlooked in such guidelines[6].

Another problem that arises with the widespread influence of chatGPT is the lack of consistency in the policy of generative AI's usage in one university. Despite the positive educational potential of Large Language Models (LLMs) like ChatGPT, their widespread availability poses a challenge for academic integrity. ChatGPT enabled student laziness due to the accessibility of answers which diminishes student's curiosity and willingness to search for knowledge. The generated work done by some students devalues the genuine work of others who authentically completed their work, complicating the assessment of students' true abilities, and providing deceptive assessment of students abilities. Our institution faces the dual task of integrating this technology to enhance learning while simultaneously preventing its exploitation for cheating. The problem is amplified by the difficulty in tracing the origins of AI-generated content, which can pass through plagiarism detection software undetected. Moreover, the current one-size-fits-all AI models do not align perfectly with our institution's specific academic standards and learning objectives. This gap necessitates the creation of a tailored LLM solution that aligns with the university's curricular demands and maintains the integrity of academic work in our unique educational setting.

5. Proposed Solution

Moving beyond chatGPT, our proposed solution is the implementation of an internal, customized chatGPT(temporarily called internalGPT for short) as a service that allows educators or faculty members of the school to build their own teaching assistant AIs. InternalGPT is built on top of chatGPT using openAI's API , while specifically uses a course-specific knowledge archive to train itself and provide students with information that is relevant to that course. The coursework's material, such as books, recorded lectures, additional resources will be provided by the lecturer to customize the model, if they choose to use the service. Students will have access to a fine-tuned to the school's data and the course's materials specifically, ensuring a context-rich experience, while removing the need to use public AI tools for education.

InternalGPT will be built using ChatGPT API and LangChain, categorized by Faith Soygazi and Damla Oguz as a library for connecting and retaining memory of all the sources to our large language model[10], while ensuring that all institutions' intellectual property and production-level private information are secured. LangChain will give our large language model the ability to take into account teacher's coursework materials, such as slides, books, video lectures(using transcript of the video) and other readings. To add another level of security to the content that will be used by the LLM to generate content,

we integrate Microsoft Presidio's identification and anonymization modules for private entities in text and images such as credit card numbers, names, locations, social security numbers, and institution's intellectual properties. It does so by utilizing different algorithms and techniques of identification such as regex(pattern recognition algorithm), NER(a machine learning model for detecting entities) and Checksum(used for pattern validation)[11].

The product usage flow will be simple. First, the students will create a prompt on the designated platform, which uses OpenAI API to generate answers, ideally about the material of a specific class, or a concept that is present in the coursework. The OpenAI API will use the material of the class through LangChain and attempt to generate the student's answer. Before sending the information to ChatGPT, the application will anonymize sensitive information, giving the app another security layer. After generating the student's answer, ChatGPT's output will be de-anonymized, returning to the students their requested information. (See Figure 1).

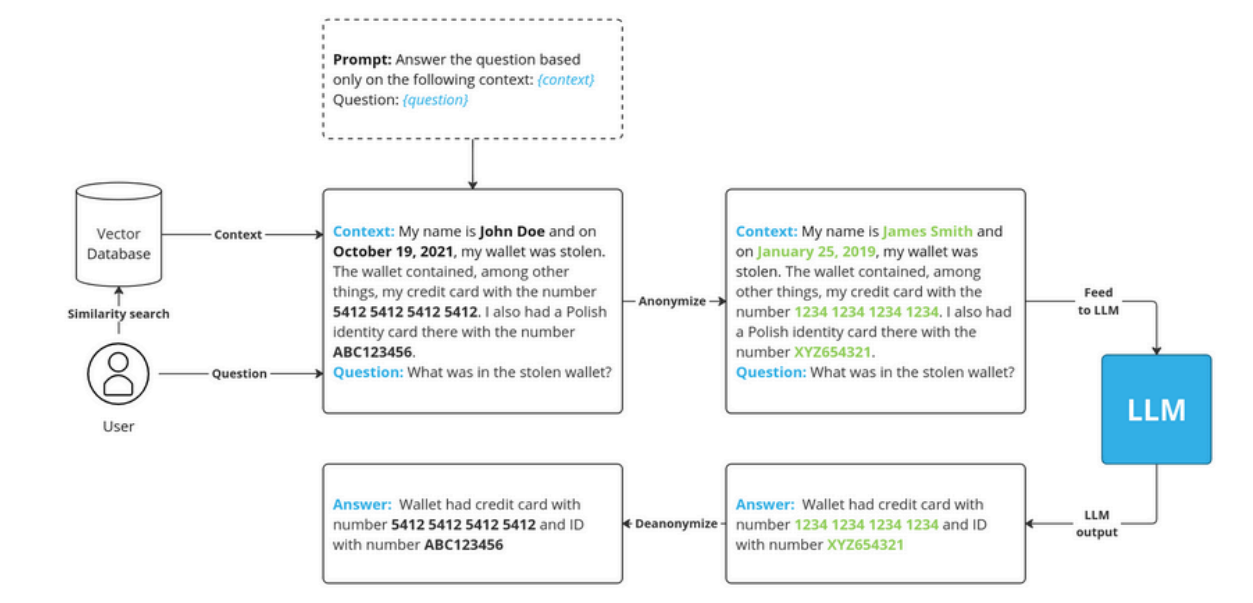


Figure 1: Product workflow (credits: Langchain Documentation)

Presidio Detection Flow

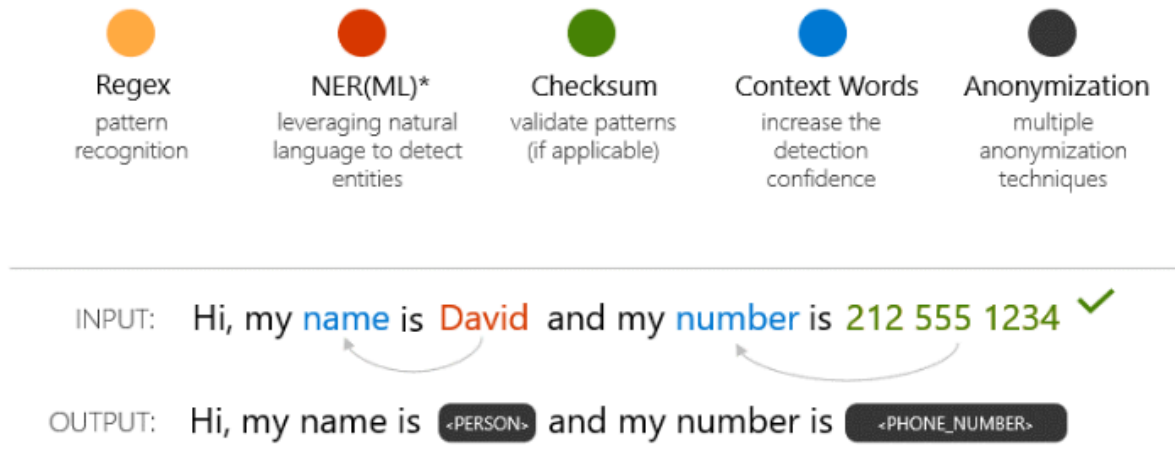


Figure 2: Microsoft Presidio's detection flow (Credits: Microsoft Presidio)

5.1 Strengths and Differentiation

Firstly, the strength of our proposed InternalGPT lies in its capability to protect the intellectual property of educational content. By operating within the infrastructure of designated universities and exclusively using institution-provided materials, InternalGPT ensures that academic content stays within the school's controlled environment, reducing the risk of unauthorized sharing or exposure of proprietary information. This is a marked advantage over the general ChatGPT model, which interacts with a wider range of data sources and users, posing a potential threat to the privacy of sensitive educational resources. InternalGPT's focused use of content is actively guarded against the common vulnerabilities present in public AI platforms. As such, InternalGPT offers a privacy-first solution for the intellectual rights of educators with no compromise to innovation.

Secondly, another strength of InternalGPT is in its exclusive, course-specific knowledge base that is meticulously curated from the actual coursework provided by the educators themselves. This novel approach to AI training ensures that the information dispensed by InternalGPT is not only accurate and unbiased but also directly relevant to the students' coursework, thereby enhancing the educational experience with context-rich interactions. While this usage of LLMs is experimental, the initial results are promising. According to T. Burns, a recent study conducted at the University of Michigan where it just implemented its own AI Assistant called U-M Maizey, using a course-specific knowledge archive, ninety percent of the questions answered by Maizey were rated as better or on par with ChatGPT. Ninety-four percent of Maizey's answers were rated good or fantastic compared to seventy-four percent for answers by ChatGPT 4.0[8]. Unlike the general ChatGPT model, which pulls from a vast but generic pool of information, InternalGPT can potentially fine-tune responses using the school's unique educational content—including textbooks, recorded lectures, and supplementary materials. This level of

customization stands in stark contrast to the broader applications of general AI models and offers a tailored educational assistant that aligns closely with the curriculum and pedagogical objectives of each course, providing a personalized learning tool that integrates seamlessly into the fabric of the school's academic ecosystem.

Lastly, beyond intellectual property protection, InternalGPT stands out for its potential to significantly reduce the workload of educators and teaching assistants. By automating responses to frequently asked student inquiries, the system can handle a substantial volume of routine questions that would otherwise require personal attention from faculty or staff. This allows educators and their assistants to focus on more complex and nuanced educational tasks, such as developing curriculum content, engaging in one-on-one tutoring, or conducting research. InternalGPT's tailored assistance extends beyond mere question-answering; it can also facilitate the grading process by pre-assessing student submissions against the coursework's knowledge base, offering preliminary feedback that educators can refine. This not only streamlines the administrative aspects of teaching but also enhances the quality of interaction between students and teachers, ensuring that human resources are optimally utilized for educational enrichment rather than repetitive tasks.

5.2 Weaknesses

InternalGPT is not without its weaknesses. The very feature that sets it apart—its customization—requires a substantial initial investment of time and resources to set up. Each course implementation demands a considerable effort from educators to supply and update the model with relevant materials, which may prove challenging alongside their regular teaching responsibilities. Additionally, the closed system of InternalGPT, while secure, means that it may not benefit from the continuous and broad learning updates of the general ChatGPT model, potentially limiting the scope of its responses to the data it was originally trained on. Moreover, there is an inherent risk that the model could propagate any biases present in the provided course materials, thus necessitating vigilant oversight and frequent reassessment of its knowledge base to ensure continued accuracy and objectivity.

This personalized and humane aspect of teaching is where InternalGPT may fall short, as AI, no matter how sophisticated, cannot replicate the genuine emotional intelligence and empathetic connection that human educators provide. The misalignment of motivation, as highlighted by Bates et al., becomes apparent when considering the dimension of emotional influence in learning. While InternalGPT can offer information and assistance tailored to a specific curriculum, it lacks the capability to adapt its interactions based on the emotional state of the student, a critical factor in the learning process[7]. The absence of this emotional engagement could potentially create a learning environment that feels more transactional than transformational, missing opportunities to inspire and motivate students through the nuanced, caring approach that characterizes the best of human teaching. As such, despite the many conveniences that InternalGPT may introduce, it is essential to acknowledge and address its limitations in fulfilling the complete spectrum of student needs that extend beyond academic instruction to include emotional and psychological support.

The future of education lies within the relationship between human educators and artificial intelligence. Educators, who are overworked averaging 53 hours per week (NCES) would opt for some incorporation of LLMs to minimize their workload [1] considering that AI can sufficiently grade papers to the standard

of the marker and can assist in most manners required to help the educator teach. The adoption of LLMs would result in the flourishing of education's intent as students would get access to personal forms of teaching, LLMs would inhibit laziness elevated by the current reliance on chatGPT outside of school, and as stated LLMs would increase the curiosity of many students encouraging learning.

6. Counter Arguments

In the current system where education is not funded sufficiently, the implementation of InternalGPT may be unrealistic due to the high cost of development. Creating and maintaining an AI system requires skilled personnel, meaning people knowledgeable in machine learning and data security. Due to the nature of InternalGPT, data is not at risk of external parties gaining access online, however the people responsible for the development of InternalGPT's maintenance must strictly adhere to data laws.

This personalized and humane aspect of teaching is where InternalGPT may fall short, as AI, no matter how sophisticated, cannot replicate the genuine emotional intelligence and empathetic connection that human educators provide. Researchers mention the misalignment of motivation between the modern teaching AI and the current educational foundation. For example, Bates et al mention that emotions can greatly influence students' learning capability, therefore, students often learn better when they feel that their teachers care about them and apply teaching with a more humanitarian approach [4]. While InternalGPT can offer information and assistance tailored to a specific curriculum, it lacks the capability to adapt its interactions based on the emotional state of the student, a critical factor in the learning process[7]. This new technology can appear frighteningly human-like, but lacks fundamental human character to resonate with students. As such, despite the many conveniences that InternalGPT may introduce, it is essential to acknowledge and address its limitations in fulfilling the complete spectrum of student needs that extend beyond academic instruction to include emotional and psychological support.

While universities are responsible for the sourcing and utilization of different articles and books, the level of control the institutions would possess with the usage of LLMs is unprecedented. The usage of customizable LLMs in educational institutions suggests that the institution themselves have complete control over the customization and the content that the LLMs use to train itself. Meaning, the institution directly controls the information that the students learn. The provided information must be regulated by an external body to prevent biased information from being spread potentially leading to the indoctrination of students. It's crucial, therefore, that educational institutions maintain transparency in their use of LLMs and ensure that the information is broad, balanced, and inclusive of diverse perspectives to foster critical thinking and a well-rounded education. Due to the power of AI, it can be dangerous under the wrong hands and inherently incapable of causing harm to anyone or any institution.

7. Conclusion

As bigger and better AI tools are released, the need for AI regulations and privacy rises with it. Our InternalGPT proposal stands as a needed solution for the ever-evolving AI in education space by combining the cutting-edge capabilities of AI with priority for privacy and intellectual property protection. It offers a new paradigm of education, finely tuned to the school's curriculum, and significantly eases the pedagogical burden on educators and assistants by shouldering the repetitive load of addressing standard inquiries and providing preliminary assessments. However, it is important to

recognize that while InternalGPT can be a tool for the betterment of the education sector as a whole, it should not be treated as a religion. The absence of emotional intelligence—a quality intrinsic to human educators—underscores a gap that technology has yet to bridge, highlighting the irreplaceable value of human interaction in the learning process.

As we move forward, it is essential to balance the deployment of InternalGPT with support for the human elements of teaching that foster a nurturing and motivational educational environment. By doing so, we ensure that the introduction of AI into education not only safeguards the privacy of content but also complements the irreplaceable human interaction that help push for academic success and learning. Ultimately, the goal is to provide an educational experience that is both intellectually robust and emotionally fulfilling, preparing students to thrive in the ever-evolving world.

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